Thesis Proposal

Topic Modeling and Thematic Analysis of JFK Assassination Files Using NLP

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1 Problem Definition

In 1963, the President of the United States of America, John F. Kennedy, was assassinated in Dallas. Many classified government records, including memos, investigation reports, and transcripts, documented this shocking and historically significant event of the Cold War era. Those records were not accessible to the public for decades. On 18 March 2025, however, the U.S. government released more than 80,000 pages of these records in what they described as the most complete public disclosure to date, containing only minimal remaining redactions limited to privacy or national-security-sensitive details (Horton, Sardarizadeh, & Wendling, 2025). This release gives researchers a unique opportunity to study the entire collection. However, the large volume and unstructured format of the documents make them difficult to read and analyze manually. To partially overcome this problem, and to set up an ML pipeline for automation, Optical Character Recognition (OCR) models together with Natural Language Processing (NLP) and topic modeling techniques will be used throughout this project.

To investigate the published documents in detail using Machine Learning (ML) techniques, two complementary methods will be applied: topic modeling and thematic analysis. Topic modeling is generally considered an unsupervised text-mining technique that detects latent structures in a large corpus by identifying groups of words that tend to co-occur across documents. These sets of words form "topics," which indicate the types of themes existing in the documents, such as mentions of specific countries, intelligence operations, key individuals, or theories about the assassination. While arguably the most popular model, Latent Dirichlet Allocation (LDA), was introduced in (Blei, Ng, & Jordan, 2003), more general overviews of topic modeling are given in (Griffiths & Steyvers, 2004). Topic modeling becomes a useful tool because it can quickly process thousands of documents and help the researcher find patterns more efficiently than manual reading.

However, topic modeling on its own may sometimes provide ambiguous or difficult-to-interpret results. Thematic analysis has therefore been added as the second approach. Thematic analysis is a qualitative method in which the researcher closely reads selected portions of the text and reflects on the meanings underlying the identified topics (Braun & Clarke, 2006). This approach introduces human judgment into the procedure in order to ensure that statistically derived topics reflect insights relevant to the historical and political context surrounding the JFK files. Combining these approaches retains the depth of human interpretation needed for meaningful and trustworthy results while gaining computational efficiency.

The main objective of this work is to determine how key themes in the JFK Assassination Files can be identified using NLP techniques and what these themes may reveal about historical events and intelligence operations related to the case. This is of great importance from both an academic and a practical perspective. For historians and researchers in the digital humanities, it provides a structured, data-driven way to explore the JFK documents, potentially uncovering new insights or confirming established patterns in how the assassination has been documented. As demonstrated by (Newman & Block, 2006), such approaches have been applied in similar academic contexts. For analysts in the field of intelligence, this project offers an example of how modern tools, such as text mining, can support the processing of large collections of declassified files, a task resembling the workflows of real intelligence analysts, as demonstrated in (Vayansky & Kumar, 2020). This thesis therefore addresses the challenge of working with large, unstructured text data by combining machine learning with human interpretation in a comprehensive approach.

2 Literature Review

This project is located within research on NLP, topic modeling, intelligence analysis, and the digital humanities. The methodological choices remain flexible at this stage, as the documents are still in the early stages of processing, particularly with respect to OCR. Nevertheless, substantial literature has been reviewed to understand the range of existing techniques and their applications in comparable contexts.

One popular method for large-scale text analysis, topic modeling, is designed to uncover hidden themes within a document collection. Among the many variants, Latent Dirichlet Allocation (LDA) is a probabilistic model that has been applied to sources ranging from historical newspapers to the scientific literature (Blei et al., 2003; Griffiths & Steyvers, 2004; Newman & Block, 2006). A more approachable overview of how models like LDA can help organize large, unstructured archives can be found in (Blei, 2012).Topic modeling has also been widely used in the digital humanities to track trends over time and uncover underrepresented narratives in historical sources (Mimno, 2012).

More recent approaches include embedding-based models such as BERTopic and Top2Vec, which rely on pre-trained language models to capture semantic similarity. These methods may prove especially useful when OCR output is noisy or disrupts traditional word co-occurrence patterns. At this stage, the approach used in this thesis remains open to experimenting with both traditional probabilistic models and more recent neural approaches, depending on their performance on the corpus.

Drawing on thematic analysis from a qualitative perspective, the approach set out in (Braun & Clarke, 2006), is adopted in this project, which emphasizes systematic coding and theme development to support or refine automatically generated topics. Integrating computational modeling with manual review is increasingly common in interdisciplinary research (Boyd-Graber, Hu, & Mimno, 2017; Roberts et al., 2014), particularly when working with historical texts where contextual nuance is essential.

Research in intelligence analysis has demonstrated how NLP can support the review of large declassified document collections. Analysts often encounter challenges similar to those in this project, including processing large volumes of unstructured data under time or resource constraints (Atallah, McDonough, Raskin, & Nirenburg, 2001). Previous work has shown how topic modeling can help identify patterns in released government documents and support historians or researchers in detecting key themes (Vayansky & Kumar, 2020).

Advanced models for topic modeling include Dynamic Topic Models (Blei & Lafferty, 2006) and Structural Topic Models (Roberts et al., 2014). These methods allow the inclusion of metadata beyond raw text, enabling researchers to analyze how topical prevalence changes over time or across document sources. The choice of model will depend on the degree of structure within the dataset and the availability of useful metadata.

In summary, the literature highlights a wide range of topic modeling and thematic analysis approaches suitable for historical and unstructured collections such as the JFK assassination records. The final methodological decisions will depend on the outcome of the OCR phase but will be guided by both traditional approaches (e.g., LDA) and recent NLP advances within a human-in-the-loop framework for interpretation.

3 Methodology

The open source JFK Assassination Records, released in March 2025 by U.S. National Archives (National Archives and Records Administration, 2025) will be used and analysed. This dataset consists of about 80,000 pages of scanned documents in PDF format. Many of these are typewritten government reports or memoranda with additional handwritten remarks, stamps and tables from the 1960s and later. As the files are not machine-readable by default, the first step involves converting them into text using Optical Character Recognition, or OCR.

3.1 Data Acquisition and OCR Validation

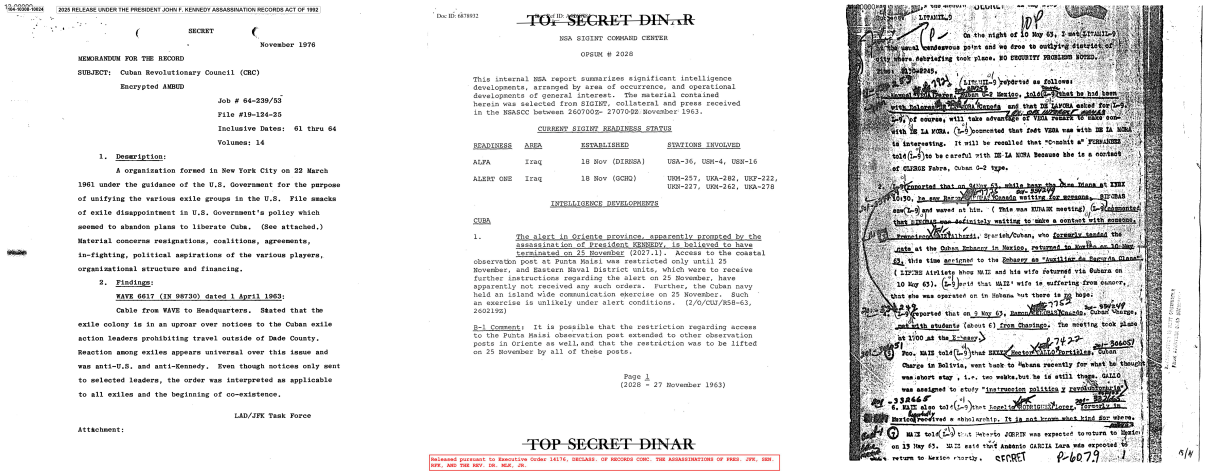
To guide this process, a set of document-level annotations was developed to assess OCR complexity. These include the following binary or categorical fields:

- includes_handwriting
- has_shadowy_background
- document_quality
- text_density
- has_stamps
- has_redactions
- has_forms
- has_tables
- is_typewritten
- paper_condition
- primary_characteristics

These fields enable the classification of documents by visual and structural difficulty and route them to the most appropriate OCR tier. Documents are given a complexity level score: simple, average, or complex, based on how many of the mentioned criteria exist in the document. In the examples in Figure 1, the first document is cited as ”simple” because it contains only a simple, easy-to-extract typewritten text. The second one includes a table, which makes it an ”average” complexity file. The last one has a very shadowy background, low quality, and lots of handwritten corrections on top, which makes it ”complex”.

In the next phase, a comparison is made with the traditional Tesseract OCR i.e., an open-source tool in general use to recognize printed text (Smith, 2007), and DeepSeek OCR, i.e., a pre-trained system that compresses images of pages into compact sets of “vision tokens” and then decodes them into structured text (Wei, Sun, & Li, 2025). In addition, different pre- and post-processing techniques were tested on sample pages that were retranscribed manually by hand. Those two models are selected since both are free and open source, unlike Google’s DocumentAI, LandingAI, and others. This gives the possibility to run them locally with reproducibility. It is also possible to fine-tune and customize them based on differences between the documents.

The accuracy of a model is calculated by taking the average of character, word, and line similarities and Levenshtein similarity, which is the “number of times you would have to insert, delete, or substitute a character from string 1 to make it like string 2.” (Coates & Breitinger, 2023). In the analysis, DeepSeek OCR achieved 75% accuracy in the sample files, while Tesseract achieved 70%. By analyzing specific cases, it is found that when the document has a shadowy background, forms, tables, and/or other features that make it harder to read, DeepSeek OCR actually performs much better than Tesseract. Meanwhile, when document



Document with "Simple" Extraction Complexity

Document with "Average" Extraction Complexity

Document with "Complex" Extraction Complexity

Figure 1: Comparison of simple, average, and complex coded documents.

quality is good and other features are absent, Tesseract, even though a simpler, less complex model, performs better in terms of text similarity with the manually extracted samples.

After both OCR models extracted text from the documents, post-processing steps were performed by means of Large Language Modeling usage, specifically a Llama model, i.e., "a collection of pretrained and instruction tuned generative models in 8B parameters" created by Meta (Grattafiori & et al., 2024), with a prompt to clean the text and get rid of small errors such as empty spaces that split words like "Ken Nedy" instead of "Kennedy". With that additional step, the accuracy of Tesseract increased by 3%, where DeepSeek OCR accuracy increased by 1.5%. Even though the difference is small in a random sample of only 10 documents, it improves performance in specific kinds of documents, which will be revealed after performing the test with a stratified sample of a larger size.

Based on these preliminary tests, the initial expectation is to use a mix of Tesseract OCR and Llama post-processing for simpler files, meanwhile using DeepSeek OCR for more complex ones. This will help to use the resources more efficiently, since DeepSeek OCR requires GPU usage, which is needed to manage carefully to avoid costs, and potentially allocate usage across several providers, such as Google Colab and AWS.

Once the text has been extracted from the OCR, the intention is to clean the output by removing line breaks, fixing broken words, correcting obvious errors, and possibly carrying out spelling correction. Later stages of work may involve standard NLP preprocessing like tokenization, lowercasing, removal of stopwords and punctuation, and lemmatization, that is, reducing words to their base form. The extraction of metadata will also be explored if available and the application of Named Entity Recognition (NER) to identify people, organizations, and locations in the documents, which are of particular importance in intelligence records (Grishman, 1997).

3.2 Topic Modeling (Planned)

Once OCRed and pre-processed, the plan is to use topic modeling in order to uncover major themes throughout the document set, but no particular model has yet been selected. One such option is LDA, Latent Dirichlet Allocation, a well-known probabilistic model that depicts documents as mixtures of topics and topics as distributions over words (Blei et al., 2003). LDA has seen numerous applications in the digital humanities and historical scholarship (Newman & Block, 2006; Mimno, 2012) and remains one of the standard baselines for unsupervised topic discovery.

Newer topic modeling methods will also be considered, like BERTopic, which can better understand word meanings and group similar documents together (Boyd-Graber et al., 2017). These models might work well even if the OCR text is not perfect. If enough extra information found, like who wrote the document or when it might be useful, Structural Topic Models may be used, which can include that information when finding topics (Roberts et al., 2014). If the documents span many years, dynamic topic models could also be tested to see how topics change over time (Blei & Lafferty, 2006). Since the project is still in the early stages, different models will be tested and compared on how well they work using common quality measures like topic coherence and perplexity (Röder, Both, & Hinneburg, 2015).

3.3 Thematic Analysis and Interpretation

Once topic clusters are generated, they will be interpreted using a qualitative approach as drawn on Braun and Clarke's 2006 methodology for thematic analysis, and a sample of documents in each topic will be read manually before assigning a descriptive label to the content. This process helps to validate the output of the model and also makes the model more readable. For instance, if one of the topic clusters features keywords such as "Castro," "Cuba," "Bay of Pigs," and "CIA," that cluster would be labeled as "Cuban Intelligence Operations."

Documentation of the labeling process will take place, as well as an examination of historical sources, to determine if the results of our computational topic analysis match existing accounts or narratives. It is anticipated that if themes correlate with topics found in official investigations/reports such as the Warren Commission, citations will be provided to these documents. If, however, there are any new or unexpected themes the significance of these items will be assessed through careful reading and contextual research. As appropriate, it will be the goal to work with a historian or subject-matter expert for independent confirmation of our findings.

3.4 Visualization and Presentation of Results

The findings will be visually presented in ways that support intuitive exploration of the topics and entities identified through the analysis. For example, topic maps or cluster diagrams might be created to provide insight into how topics are related to one another, or word clouds or bar charts displaying some of the most frequent terms in a given topic. When time information is part of the data, timelines can also be created showing how the importance of certain themes evolved over the years. Finally, possibly network graphs will be used showing the interrelatedness of key named entities, such as Lee Harvey Oswald, government agencies, or geographic locations, across documents. To interactively explore the structure of the topic models, visualization tools such as LDAvis may be employed that provide an easy-to-use way to examine the

relevance and overlap of topic terms (Sievert & Shirley, 2014). This would facilitate communicating our results more effectively, specifically to those users who do not have a deep technical understanding of NLP.

4 Conclusion

The workflows are currently still a work in progress and continue to be refined at this point in time with regard to the initial phase of the implementation of OCR processing. As of today, multiple OCR systems are currently being used, most notably Tesseract and DeepSeek OCR, while continually refining and modifying both the pre- and post-processing methods based on how complicated a document may be. Based on these test findings, the best OCR engine(s) will be determined to use for specific document extensions and the anticipated reliability of text extraction that is to be utilised for downstream processing of the data. This flexible and iterative operational model gives an opportunity to modify workflows in relation to the practicalities of dealing with historical scans while still maintaining a keen awareness of our analytical objectives.

The end result will be a thematic overview of the JFK documents grounded in computational analysis and human interpretation. This work aims not only aids the academic literature in digital humanities and intelligence analysis but also serves to provide a helpful roadmap for researchers, historians, and the public interested in navigating this vast and long-awaited archive.

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